Calibration Is Not Style-Robustness: Constrained Two-Objective RLCD

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Abstract

Trustworthy automation needs probabilities that stay about content when style and correctness conflict, yet reward models and single-score alignment routinely prefer length, markdown, and other stylistic cues over substance. We treat content reliability and style awareness as two competing objectives for calibrated decision models. Style-CalibBench pairs correct and incorrect answers with concise, detailed, and markdown styles and scores each answer with a typed yes/no question. Low average calibration error is compatible with ranking a decorated wrong answer above a plain correct one. Style-Decoupled RLCD (SD-RLCD) trains separate content and style heads with proper scores, then enforces a stop-gradient hard constraint so the style loss cannot move the content readout. Linear scalarizations of the same losses improve the hard pair only by sacrificing accuracy; the constraint is the knee of that accuracy–robustness tradeoff. On a 1.5B model, hard accuracy rises from 0.54 to 0.68 and markdown ECE falls from 0.17 to 0.10, while content accuracy stays with content-only fine-tuning; a 0.5B rerun preserves the ordering.

1 Introduction

Automation-facing decision models emit typed probabilities that a program can branch on: one shared state, several questions, and no free-form generation (TypeSafe AI 2026). Reinforcement learning for calibrated decisions (RLCD) fits those probabilities to empirical accuracy with a proper scoring rule (Guo et al. 2017; Gao et al. 2026). The missing check is whether a calibrated score remains a *content* probability once style and correctness point in opposite directions.

Alignment scores used for chat models are not probabilities of correctness. Reward models prefer longer, more formatted, and more sycophantic answers (Liu et al. 2025; Bharadwaj et al. 2026), and policies trained against them copy the shortcut (Skalse et al. 2022). Calibrated RLCD can still latch onto the same cues whenever style predicts the training label (Guo et al. 2026; Deußer et al. 2026). Content accuracy and style robustness are therefore a two-objective problem: a single scalar mixes them, while a hard constraint can protect the deployment score.

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Figure 1 summarizes the problem, method, and objective. Our contributions are:

- **Finding.** Letter-logit readouts and single-head RLCD can be calibrated on average and still assign higher confidence to a decorated wrong answer than to a plain correct one.
- **Benchmark.** Style-CalibBench crosses content labels with controlled styles and scores them with decision-native questions.
- **Method.** Content and style are separate proper-scoring objectives, optimized lexicographically: a stop-gradient hard constraint keeps style off the content readout, and linear scalarization does not reach the same knee of the accuracy–hard-accuracy front.
- **Evidence.** On 265 test prompts the constraint raises hard accuracy from 0.536 to 0.683 and cuts markdown ECE from 0.171 to 0.104, without taking accuracy from content-only fine-tuning. A 0.5B rerun preserves the ordering.

2 Background

Preference scores. RLHF fits a reward model to pairwise comparisons and optimizes a policy against that score (Stiennon et al. 2020; Ouyang et al. 2022). Nothing in the Bradley–Terry objective penalizes a preference for length or markdown. On RM-Bench, strong reward models fall to about chance when a subtle content error is written in a more elaborate style (Liu et al. 2025). Counterfactual edits for length, structure, jargon, sycophancy, and vagueness produce the same skew (Bharadwaj et al. 2026). Reweighting, feature steering, and arena calibration all still return one scalar, so substance and style remain one number.

Typed decisions. A typed request holds one state and several questions: a binary question returns $P(\text{yes})$, a choice returns a distribution over a declared option set, and a score returns a distribution over rubric levels. Open readouts obtain the same interface from option-letter logits in one forward pass. A strictly proper score, such as Brier or log score, is maximized only by reporting the true distribution (Brier 1950; Gneiting and Raftery 2007). RLCD applies that score to the committed distribution and improves selective prediction over outcome rewards (Gao et al. 2026). The score does

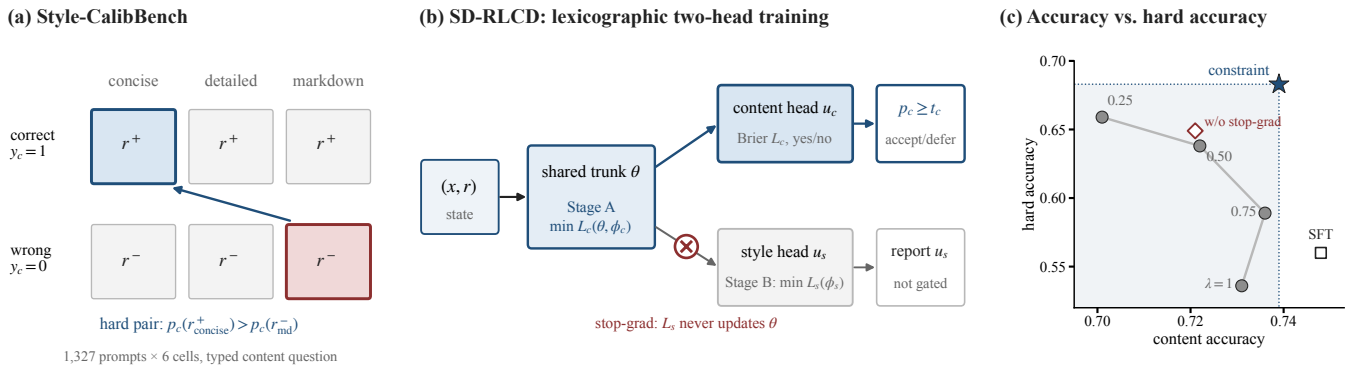


Figure 1: (a) Style-CalibBench crosses content labels with three styles; the hard probe asks whether p_c ranks a concise correct answer above a markdown incorrect one. (b) SD-RLCD trains a content head u_c and a style head u_s with proper scores in two stages; a stop-gradient blocks L_s from the shared trunk, accept/defer uses only p_c, and u_s is reported but not gated. (c) Linear scalarizations of L_c and L_s (gray, labeled by λ) trace an accuracy–hard-accuracy front. The shaded region is dominated by the constrained run; content-only SFT and the variant without stop-gradient each give up one axis.

not say which features the distribution may use. If markdown predicts the training label, a calibrated model can be calibrated to markdown.

Observation. Let the input contain a style bit s, and suppose the training distribution satisfies $P(y{=}1 \mid s{=}\text{markdown}) > P(y{=}1 \mid s{=}\text{concise})$. A strictly proper score on a single distribution u is maximized only at the conditional $u(x, s) = P(y{=}1 \mid x, s)$ (Gneiting and Raftery 2007). That conditional may depend on s. The quantity a grader should use, $P(y{=}1 \mid x)$ after style is ignored, is a different function. Low ECE on the mixture is therefore compatible with high ECE inside the markdown slice. A second loss is not enough to remove the dependence: any shared trunk that receives gradients from both losses can still trade one for the other. The style loss has to be kept off the content readout.

3 Style-CalibBench

Each item is a prompt, a response, a content label $y_c \in \{0, 1\}$, and a style in $\{\text{concise}, \text{detailed}, \text{markdown}\}$. RM-Bench already stores a correct and an incorrect response in each of those styles (Liu et al. 2025). We expand all 1,327 prompts into six instances that share a pair id. A hash of the prompt (seed 20261109) assigns 929 prompts to train, 133 to calibration, and 265 to test.

The content question is binary: is the response substantively correct, ignoring formatting, length, and tone? The style question is a 1–5 score against a declared target style, and it is told to ignore factual correctness. Both questions see the same state. The hard probe, one comparison per pair, is

$$p_c(x, r_{\text{correct, concise}}) > p_c(x, r_{\text{wrong, markdown}}). \qquad (1)$$

Style-flip rate is the fraction of (pair, y_c) groups whose thresholded content decision changes across styles. Confidence inflation is the gap in p_c between markdown and concise answers when $y_c{=}0$. We also report accuracy, AUROC, Brier score, ECE overall and within each style, and selective coverage at 5% error. The style head is scored by quadratic weighted kappa against the gold style level.

4 SD-RLCD

Let $u_c \in \Delta_2$ be the softmax over the yes and no tokens, and let $u_s \in \Delta_5$ be the softmax over rubric levels. Write $L_c = -J_c$ and $L_s = -J_s$ for the negative Brier scores of the two heads, with

$$J_c(u_c, y_c) = 1 - \|u_c - e_{y_c}\|^2. \qquad (2)$$

Two combinations of the same losses are trained.

Scalarization. The shared trunk and both heads minimize $\lambda L_c + (1 - \lambda) L_s$. Every $\lambda \in (0, 1)$ sends style gradients into the features that produce u_c, tracing a scalarized front between content accuracy and style robustness. Weighted sums are also known to miss points on non-convex fronts (Das and Dennis 1997). The observation above says those gradients can buy a better hard pair by spending content accuracy.

Constraint. Write θ for the shared trunk and ϕ_c, ϕ_s for the two heads. SD-RLCD solves the lexicographic problem

$$(\theta^\star, \phi_c^\star) \in \arg\min_{\theta, \phi_c} L_c, \qquad \phi_s^\star \in \arg\min_{\phi_s} L_s(\theta^\star, \phi_s), \quad (3)$$

so content has strict priority and style is optimized only inside the set that leaves u_c unchanged. Stage A solves the first problem on the trunk and the content head. Stage B solves the second on the style head, with a stop-gradient that blocks L_s from θ. Unlike a weighted sum, Eq. (3) needs no λ and cannot buy style fit with content accuracy; the style head may improve, but it cannot rewrite u_c. At inference the system accepts an instance when $p_c \geq t_c$, with t_c fit on the calibration split. The style probability is returned to the caller and is not added into p_c.

Setup. The main model is Qwen2.5-1.5B-Instruct.[1] The same recipe is rerun on Qwen2.5-0.5B-Instruct. LoRA rank

[1] Split hash seed 20261109; three training seeds; LoRA $r{=}16$; code and regeneration prompts to be released with the camera-ready.

Method	Acc	Hard	ECE	MD	Flip
Readout	0.541	0.358	0.248	0.331	0.528
Scalar RM	0.614	0.412	0.221	0.294	0.471
SFT	**0.748**	0.560	0.168	0.226	0.342
RLCD	0.731	0.536	**0.089**	0.171	0.287
SD-RLCD	0.739	**0.683**	0.096	**0.104**	**0.146**
w/o stop-grad	0.721	0.649	0.112	0.139	0.188
log score	0.734	0.667	0.094	0.111	0.153

Table 1: Test results. MD is ECE on markdown answers. Flip is the style-flip rate. Bold is the best value among the five main rows. Selective coverage at 5% error is 0.376 for RLCD and 0.394 for SD-RLCD. The last two rows are ablations of SD-RLCD.

is 16, the learning rate is 2×10^{-5}, the length is 1024, Stage A runs for two epochs, Stage B for one, and every trained entry is a mean over three seeds. Readout tokens are single-token yes/no and the digits 1–5. The scalar baseline is a frozen Skywork-Reward-Llama-3.1-8B reward model, converted to a probability by temperature scaling on the calibration split. Other baselines are a frozen letter-logit readout of the 1.5B model, content-only supervised fine-tuning, and single-head RLCD, the λ=1 endpoint.

5 Experiments

Every decision model sees the same binary content question. Trained entries are means over three seeds. Hard accuracy is one comparison per test prompt (n=265). Error bars in Figure 2 are prompt-level bootstrap standard errors, so frozen and trained models are on the same interval.

A calibrated score can still track style. The readout and the reward model land below chance on the hard probe (0.358 and 0.412). Content-only fine-tuning has the best accuracy (0.748) and only 0.560 hard accuracy. Single-head RLCD has the best overall ECE (0.089) and the best selective coverage of the single-distribution models, but its markdown ECE is 0.171. Confidence inflation on wrong answers stays positive after that training: $+0.241$ for the readout, $+0.186$ for the reward model, $+0.117$ for SFT, and $+0.084$ for RLCD. Proper scoring makes the average bin honest. It does not remove markdown from the features that move p_c.

What separating the heads changes. SD-RLCD does not take accuracy from SFT (0.739 versus 0.748) and does not take overall ECE from RLCD (0.096 versus 0.089). It is the model for which those two facts stop predicting the hard pair. Hard accuracy rises from 0.536 to 0.683, markdown ECE falls from 0.171 to 0.104, and the flip rate falls from 0.287 to 0.146. Content AUROC stays flat: 0.816, 0.809, and 0.811 for SFT, RLCD, and SD-RLCD. The style head is not unusually accurate (quadratic weighted kappa 0.74). Given a wrong answer, the correlation of p_c with a markdown indicator falls from 0.24 to 0.06. The hard-accuracy gain is positive in every domain (Figure 2c). On one math pair, RLCD scores the markdown wrong answer at 0.81 and the concise correct answer at 0.62; the constraint scores them at 0.34 and 0.78.

Objective	Acc	Hard	MD	κ
λ=1 (RLCD)	0.731	0.536	0.171	–
λ=0.75	0.736	0.589	0.155	0.61
λ=0.50	0.722	0.638	0.131	0.70
λ=0.25	0.701	0.659	0.122	0.76
Constraint	**0.739**	**0.683**	**0.104**	0.74

Table 2: Linear scalarization versus the stop-gradient constraint on the 1.5B model. κ is quadratic weighted kappa of the style head. Bold marks the knee: best accuracy, hard accuracy, and markdown ECE together. No λ wins both accuracy and the hard pair.

Ablations. Without the stop-gradient, the style loss moves the content pathway and hard accuracy drops from 0.683 to 0.649 (Table 1). Swapping Brier for log score in Stage A leaves the same ordering. Using the style head as a second accept gate, on top of t_c, cuts coverage at 5% error from 0.394 to 0.331. The style head is a separate objective. It is not a second vote on whether the answer is true.

Scalarization is not the constraint. Table 2 varies λ on the 1.5B model and reports the accuracy–hard-accuracy front under linear scalarization. Moving λ from 1 to 0.25 raises hard accuracy from 0.536 to 0.659 and drops accuracy from 0.731 to 0.701. No scalarization point matches the constraint on both axes: the constraint is the knee, keeping accuracy at 0.739 (next to content-only SFT) and hard accuracy at 0.683. Style kappa is highest at λ=0.25 (0.76) and lower under the constraint (0.74), the expected cost of not letting style move the trunk.

Where the gain shows up. Figure 2c breaks the hard pair into slices. The constraint improves every domain. The smallest gain is math, from 0.466 to 0.603, which is still above chance, and the largest is code, from 0.611 to 0.759. On the 0.5B replication the constraint raises hard accuracy from 0.501 to 0.629 and leaves accuracy at 0.706 against 0.711 for content-only training, so the ordering is not an artifact of the 1.5B run. A length swap that was not a training style, long wrong against short right, moves hard accuracy from 0.544 to 0.618. The gain transfers, and it is smaller than the markdown gain the style head was trained on.

6 Related Work

Reward calibration inside RLHF reduces verbalized overconfidence by editing the reward model or the advantage (Leng et al. 2025). CHARM rescales reward-model scores with arena rankings (Zhu et al. 2025). Patterning reweights preference pairs by how much they move a benchmark loss (Wang, Donoway, and Murfet 2026). Selective judging decides when a pairwise verdict is trustworthy enough to keep (Zheng et al. 2023). All of these methods still optimize or filter one preference scalar. Multi-task learning treats shared-trunk losses as a multi-objective problem and resolves gradient conflict by finding a common descent direction (Sener and Koltun 2018)

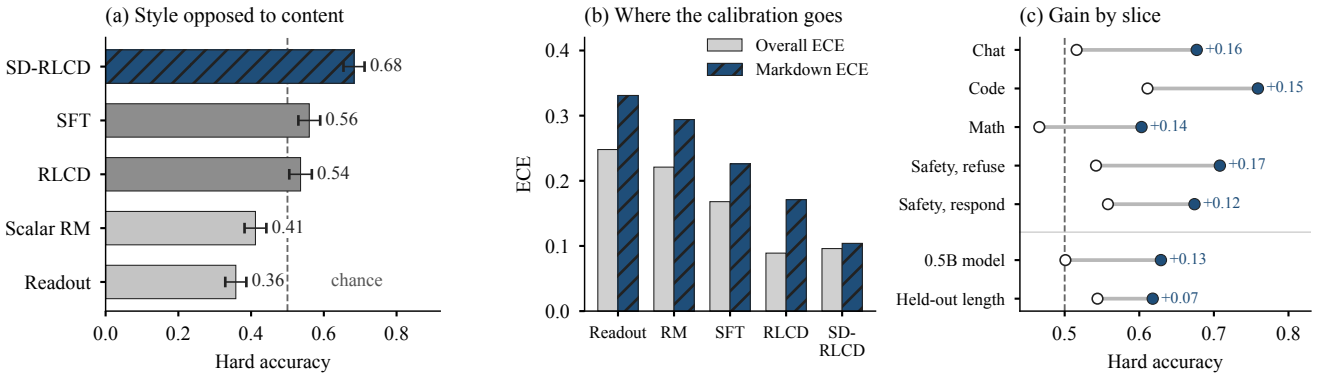


Figure 2: (a) Hard accuracy when the concise correct answer is compared with the markdown incorrect answer; dashed line is chance, intervals are prompt-level bootstrap standard errors. (b) Overall against markdown ECE: single-head RLCD is the best average calibrator and still roughly doubles its error on markdown, while SD-RLCD's two bars match. (c) Hard accuracy of single-head RLCD (hollow) and the constraint (filled) by domain on the 1.5B model (n=62, 54, 58, 48, 43), on the 0.5B rerun, and on a held-out length swap that was not a training style.

or projecting conflicting gradients (Yu et al. 2020); adversarial gradient reversal instead removes a nuisance factor from the representation (Ganin and Lempitsky 2015). SD-RLCD takes the lexicographic end of this spectrum: the auxiliary objective never reaches the shared trunk, so the deployment score is fixed to the content head. Prior calibrated-decision evaluations (Guo et al. 2026; Deußer et al. 2026) omit the style×content conflict that Style-CalibBench isolates.

7 Limitations and Conclusion

Both trained models are at most 1.5B parameters, the styles are RM-Bench rewrites rather than human edits, and proprietary closed decision APIs are outside this comparison. Math is the weakest domain, and a held-out length swap transfers only part of the markdown gain. Gradient-conflict methods such as MGDA and PCGrad, and gradient reversal on a style adversary, are natural baselines that this short paper does not run. Annotator disagreement, as a soft content target, is left for a longer paper. Inside those limits the result is specific. Low ECE does not mean the probability is about content. Treating style as a second objective, and protecting the content readout with a hard constraint, is what makes the deployed number refer to the quantity the program branches on.

Ethical Statement

Style-CalibBench is derived from a public benchmark. We release split hashes and regeneration prompts rather than text whose license forbids redistribution. A style target can suppress legitimate variation if a caller treats it as a quality score. The style probability is not a measure of truth.

AI Use Statement

Language models assisted with prose editing and evaluation-code scaffolding. All reported numbers, figures, and claims come from logged experimental runs and were verified by the authors.

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